PROBLEMAS RESUELTOS TEMAS 4, 5 y 6

TEMA 4

Diseño 1

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity Tema4_problema1 is
   Port ( EN : in  STD_LOGIC;
         E : in  STD_LOGIC_VECTOR (1 downto 0);
         Y : out  STD_LOGIC_VECTOR (3 downto 0));
end Tema4_problema1;
architecture Behavioral of Tema4_problema1 is
begin
Y<=        "1111" when EN='1' else
           "1110" when E="00" else
           "1101" when E="01" else
           "1011" when E="10" else
           "0111";
end Behavioral;
```

Diseño 2

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity Tema4_problema2 is
   generic (n: integer :=8);
   Port ( EN : in  STD_LOGIC;
         A,B : in  STD_LOGIC_VECTOR (n-1 downto 0);
         A_mayor_B, A_igual_B, A_menor_B : out  STD_LOGIC);
end Tema4_problema2;
architecture Behavioral of Tema4_problema2 is
begin
A_mayor_B<= '1' when EN='1' and A>B else '0';
A_igual_B<= '1' when EN='1' and A=B else '0';
```

```
A_menor_B<= '1' when EN='1' and A<B else '0';
end Behavioral;
```

<u>Diseño 3 hecho en clase</u>

<u>Diseño 4</u>

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity tema4_problema4 is
generic (n: integer :=8);
   Port ( A, B, C : in  STD_LOGIC_VECTOR (n-1 downto 0);
          mayor, menor : out  STD_LOGIC);
end tema4_problema4;

architecture Behavioral of tema4_problema4 is
begin
mayor<='1' when A>B and B>C else '0';
menor<='1' when A<B and B<C else '0';
end Behavioral;
```

<u>Diseño 5</u>

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity Tema4_problema5 is
   generic (n: integer :=8);
   Port ( E : in  STD_LOGIC_VECTOR (n-1 downto 0);
            sel: in std_logic_vector (2 downto 0);
            Y : out  STD_LOGIC_vector (n-1 downto 0));
end Tema4_problema5;
architecture Behavioral of Tema4_problema5 is
begin
with sel select
```

```
Y<= E when "000",
      E(n-2 downto 0) & '0'  when "001",
      E(n-3 downto 0) & "00" when "010",
      E(n-4 downto 0) & "000" when "011",
      E(n-5 downto 0) & X"0" when "100",
      E(n-6 downto 0) & '0' & X"0" when "101",
     E(n-7 downto 0) & "00" & X"0" when "110",
     E(n-8 downto 0) & "000" & X"0";

end Behavioral;
```

Diseño 6

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.std_logic_unsigned.all;

entity Tema4_problema6a is
generic (n: integer:=4);
   Port ( A : in  STD_LOGIC_VECTOR (n-1 downto 0);
          B : in  STD_LOGIC_VECTOR (n-1 downto 0);
          S : out  STD_LOGIC_VECTOR (n-1 downto 0);
          Cout : out  STD_LOGIC);
end Tema4_problema6a;

architecture Behavioral of Tema4_problema6a is
suma: std_logic_vector (n downto 0);
begin
suma<= ('0'& A) + ('0' & B);
s<=suma (n-1 downto 0);
Cout <= suma (n);
 end Behavioral;
```

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.numeric_std.all;

entity Tema4_problema6b is
generic (n: integer:=4);
   Port ( A : in  STD_LOGIC_VECTOR (n-1 downto 0);
          B : in  STD_LOGIC_VECTOR (n-1 downto 0);
          S : out  STD_LOGIC_VECTOR (n-1 downto 0);
          Cout : out  STD_LOGIC);
end Tema4_problema6b;

architecture Behavioral of Tema4_problema6b is
suma: unsigned (n downto 0);
begin

suma<= ('0'& unsigned(A)) + ('0' & unsigned (B));
s<=std_logic_vector(suma (n-1 downto 0));
Cout <= suma (n);
end Behavioral;
```

Diseño 7

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.std_logic_unsigned.all;

entity Tema4_problema7 is
generic (n: integer:=4);
   Port ( A : in  STD_LOGIC_VECTOR (n-1 downto 0);
           B : in  STD_LOGIC_VECTOR (n-1 downto 0);
```

```
        Sel: in std_logic;
        Res : out  STD_LOGIC_VECTOR (n-1 downto 0);
        Cout : out  STD_LOGIC;
       Over: out std_logic);
end Tema4_problema7;

architecture Behavioral of Tema4_problema7 is
suma: std_logic_vector (n downto 0);
BC: std_logic_vector (n-1 downto 0);
begin
BC<=not B when sel='1' else B;
suma<= ('0'& A) + ('0' & BC)+sel;
res<=suma (n-1 downto 0);
Cout <= suma (n);
Over<= '1' when A(n-1)=B(n-1) and A(n-1)/=suma(n-1) else '0';
```

<u>Diseño 8</u>

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.numeric_std.all;

entity Tema4_problema8 is
generic (n: integer:=4);
   Port ( A : in  STD_LOGIC_VECTOR (n-1 downto 0);
        B : in  STD_LOGIC_VECTOR (n-1 downto 0);
        R : out  STD_LOGIC_VECTOR (2*N+1-1 downto 0));
end Tema4_problema8;

architecture Behavioral of Tema4_problema8 is
prod_AB: unsigned (2*N+1-1 downto 0);
prod_3A: unsigned (2*N+1-1 downto 0);
```

```
prod_5B: unsigned (2*N+1-1 downto 0);
suma: unsigned (2*N+1-1 downto 0);
begin
prod_AB<= resize((unsigned(A)) * (unsigned (B)),2*N+1);
prod_3A<= resize((unsigned(A)) * 3,2*N+1);
prod_5B<= resize(5 * (unsigned (B)),2*N+1);
suma<=prod_AB + prod_3A + prod_5B;
Res <= std_logic_vector(suma);
end Behavioral;
```

Res <= std_logic_vector (prod_AB + ... + prod_5B);

Diseño 9

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;
use IEEE.numeric_std.all;

entity Tema4_problema9 is
   Port ( A : in  STD_LOGIC_VECTOR (3 downto 0);
         CE : in  STD_LOGIC;
         D1 : out  STD_LOGIC_VECTOR (7 downto 0);
         D2 : out  STD_LOGIC_VECTOR (7 downto 0));
end Tema4_problema9;

architecture Behavioral of Tema4_problema9 is
type matriz is array (0 to 15) of std_logic_vector (7 downto 0);
constant
ROM: matriz := (X"00",X"03",X"45",X"EF",X""55",X"2C",
X"79",X"AD",X"87",X"44",X"B9",X"AA",X"FF",X"58",X"9D",X"66");
begin
D1<=ROM(Conv_integer(A)) when CE='0' else (others=>'Z');
D2<=ROM(To_integer(unsigned(A))) when CE='0' else (others=>'Z');
```

```
end Behavioral;
```

Diseño 10 Ej exámen → Con numeric y/o unsigned

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.numeric_std.all;

entity Tema4_problema10 is
generic (m: integer:=2; n: integer:=4);
   Port ( sel : in  STD_LOGIC_VECTOR (2 downto 0);
          Des : in  STD_LOGIC_VECTOR (n-1 downto 0);
          A : in  STD_LOGIC_VECTOR (n-1 downto 0);
          B : in  STD_LOGIC_VECTOR (n-1 downto 0);
          R: out std_logic_vector (n-1 downto 0)  );
end Tema4_problema10;

architecture Behavioral of Tema4_problema10 is
begin
with sel select   --asignación seleccionada de señal
R<= A and B when "000",
     A xor B when "001",
 std_logic_vector(unsigned(A)+ unsigned(B)) when "010",   IEEE.std_logic.unsigned (std_logic_vector) R<=A+B;
 std_logic_vector(unsigned(A)- unsigned(B)) when "011",
std_logic_vector(unsigned(A)+1) when "100",
 std_logic_vector(unsigned(A)-1) when "101",
std_logic_vector(shift_right(unsigned(A),to_integer(unsigned(Des)))) when "110",
 std_logic_vector(shift_left(unsigned(A),to_integer(unsigned(Des)))) when others;

end Behavioral;
```

```
R<= '0' & A(n-1 downto 1);
R<= to_std_logic_vector (to_bitvector (A) srl 1);

R<= A(n-2 downto 0) & '0';
R<= to_std_logic_vector (to_bitvector (A) sll 1);

std_logic_vector (rotate_right (unsigned(A), to_integer (unsigned(Des))))
R<= A(0) & A(n-1 downto 1);
R<= to_std_logic_vector (to_bitvector (A) ror 1);

std_logic_vector (rotate_left (unsigned(A), to_integer (unsigned(Des))))
R<= A(n-2 downto 0) & A(0);
R<= to_std_logic_vector (to_bitvector (A) rol 1);
```

TEMA 5

Diseño 1

```
library IEEE;
```

```
use IEEE.STD_LOGIC_1164.ALL;


entity Tema5_problema1 is
generic(n: integer:=4);
   Port ( Din : in  STD_LOGIC_VECTOR (n-1 downto 0);
          OE : in  STD_LOGIC;
          Dout : out  STD_LOGIC_VECTOR (n-1 downto 0));
end Tema5_problema1;


architecture Behavioral of Tema5_problema1 is


begin
process (Din, OE)
begin
if OE='0' then
Dout<=Din;
else Dout<=(others=>'Z');
end if;
end Behavioral;
```

Diseño 2

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;


entity tema4_problema2 is
 generic (n: integer :=4);
   Port ( A : in  STD_LOGIC_VECTOR (n-1 downto 0);
            B: in  STD_LOGIC_VECTOR (n-1 downto 0);
            Cin: in std_logic;
            Cout: out std_logic;
            S : out  STD_LOGIC_vector (n-1 downto 0));
end tema4_problema2;
```

$S = a \oplus b \oplus Cin$

$Cout = ab + (a+b)Cin$

$C(0) = Cin$

$$
\begin{array}{cccc}
 & C(3) & C(2) & C(1) \\
A_3 & A_2 & A_1 & A_0 \\
B_3 & B_2 & B_1 & B_0 \\
\hline
\downarrow & S_2 & S_1 & S_0
\end{array}
$$

$Cout \Leftarrow C(4)$

$C = n+1$ bit

```
architecture Behavioral of tema4_problema2 is
begin
process (A, B, Cin)
variable C: std_logic_vector(n downto 0);
begin
C(0):=Cin;
for i in 0 to n-1 loop
S(i)<= A(i) xor B(i) xor C(i);
C(i+1):= (A(i) and B(i)) or ((A(i) xor B(i)) and C(i));
end loop;
Cout<=C(n);
end process;
end Behavioral;
```

Diseño 3

Si En = '1' then
 Q ⇐ D_1
else

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Tema5_problema3 is
generic(n: integer:=4);
   Port ( D : in  STD_LOGIC_VECTOR (n-1 downto 0);
          E : in  STD_LOGIC;
          Q : out  STD_LOGIC_VECTOR (n-1 downto 0));
end Tema5_problema3;

architecture Behavioral of Tema5_problema3 is
begin
process (D, E)
begin
```

```
if E='1' then
Q<=D;
end if;
end process;
end Behavioral;
```

Diseño 4 * Exámen sin ESDD y ESDI

ESDD n-1 ... 1 0

n-1 ... 1 0 ESDI

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity tema5_problema4 is
 generic (n: integer :=4);
    Port ( D : in  STD_LOGIC_VECTOR (n-1 downto 0);--entrada de datos carga paralela
           clk: in std_logic;
           clear: in std_logic;
           ESDD: in std_logic;
           ESDI: in std_logic;
            sel: in std_logic_vector (1 downto 0);
            Q : out  STD_LOGIC_vector (n-1 downto 0));
end tema5_problema4;

architecture Behavioral of tema5_problema4 is
signal registro: std_logic_vector (n-1 downto 0);
begin
process (clk, clear)
begin
if clear='0' then
 registro <= (others=>'0');
 elsif rising_edge(clk) then
 case sel is
```

```
when "01"=>
'0'
registro<= ESDD & registro (n-1 downto 1);
std_logic_vector (shift_right(unsigned(registro), 1));   ← numeric
when "10" =>
std_logic_vector (to_bitvector(registro) srl 1);
registro <= registro(n-2 downto 0) & ESDI;
'0'
when "11" =>
registro<=D;
when others =>
registro<=registro;
end case;
end if;
end process;
Q<=registro;
end Behavioral;
```

Diseño 5

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Tema5_problema5 is
generic(n: integer:=4);
   Port ( A, B, C : in STD_LOGIC_VECTOR (n-1 downto 0);
         Y : out  STD_LOGIC);
end Tema5_problema5;

architecture Behavioral of Tema5_problema5 is
begin
process (A, B, C)
begin
if A>B and A>C then
Y<=A;
```

```
elsif B>A and B>C then
Y<=B;
elsif C>A and C>B then
Y<=C;
else --los tres numeros son iguales
                    '0'
Y<= (others=>'Z');
end if;
end process;
end Behavioral;
```

Diseño 6

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_unsigned.ALL;
--use IEEE.numeric_std.all;

entity Tema5_problema6 is
generic(n: integer:=4; m: integer:=8);   SRAM de 2^n palabras de m bits
   Port ( A: in STD_LOGIC_VECTOR (n-1 downto 0);
          Din: in STD_LOGIC_VECTOR (m-1 downto 0);
          clk, CS, WR : in  STD_LOGIC;
           Dout: out std_logic_vector (m-1 downto 0));
end Tema5_problema6;

architecture Behavioral of Tema5_problema6 is
                             0 to 2**n-1
type matriz is array (integer range 0 to 2**n-1) of std_logic_vector (m-1 downto 0);
signal memoria: matriz;
begin
process (clk, CS, WR)
```

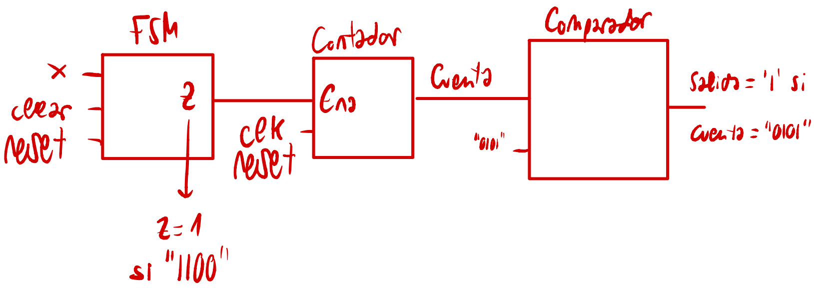
```
begin
if CS='1' then
        if rising_edge (clk) then
                if WR='1' then
                memoria (conv_integer(A))<=Din; -unsigned
                --memoria(to_integer(unsigned(A)))<=Din; - numeric
                else
                Dout<=memoria(conv_integer(A));  -unsigned
                --Dout<=memoria(to_integer(unsigned(A))); - numeric
          end if;
        end if;
else
Dout<= (others=>'0');
end if;
end process;
end Behavioral;
```

<u>Diseño 7</u>

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_unsigned.ALL;
--use IEEE.numeric_std.all;
entity Tema5_problema7 is
generic(n: integer:=4);
   Port (clk,Ena, Load, clear,sel: in STD_LOGIC;
        E: in STD_LOGIC_VECTOR (n-1 downto 0);
         Q: out std_logic_vector (n-1 downto 0);
        Carry: out std_logic);
end Tema5_problema7;
```

```
architecture Behavioral of Tema5_problema7 is

signal cuenta: std_logic_vector(n-1 downto 0);
constant fin_cuentaasc:std_logic_vector(n-1 downto 0):=(others=>'1');
constant fin_cuentadesc:std_logic_vector(n-1 downto 0):=(others=>'0');

begin
process (clk, Clear)
begin
if Clear='1' then
        cuenta<=(others=>'0');
elsif rising_edge (clk) then
        if Load='1' then
        cuenta<=E;
        elsif Ena='1' then    Entrada de habilitación
                case sel is
                when '0'=> cuenta<=cuenta+1;--con el paquete unsigned
                --when '0'=> cuenta<=std_logic_vector(unsigned(cuenta)+1); con numerico
                when '1'=> cuenta<=cuenta-1;--con unsigned
                --when '1'=> cuenta<=std_logic_vector(unsigned(cuenta)-1); con numerico
         end case;
        end if;
end if;
end process;

process (cuenta,Load,Ena,sel)
begin
carry<='0';
if Ena='1' and load='0' and clear='0' then
case sel is
                when '0'=>  if cuenta=fin_cuentaasc then
```

```
                                        carry<='1';
                                        end if;
                    when '1'=> if cuenta=fin_cuentadesc then
                                        carry<='1';
                                        end if;
end case;
end if;
end process;
Q<=cuenta;
end Behavioral;
```



TEMA 6

Diseño 1

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Tema6_problema1 is
  Port ( clk : in  STD_LOGIC;
       reset : in  STD_LOGIC;
       ent_ser : in  STD_LOGIC;
       det : out  STD_LOGIC);
end Tema6_problema1;

architecture Behavioral of Tema6_problema1 is
type estados is (Inicio, S_1, S_10, S_101, S_1011, S_10110);
signal estado_actual, estado_siguiente: estados;
begin
process (clk,reset)
begin
if reset='0' then
estado_actual<=inicio;
elsif rising_edge (clk) then
```

```
estado_actual<=estado_siguiente;
end if;
end process;
process (ent_ser,estado_actual)
begin
Det<='0';

case Estado_actual is

when Inicio=>
 if Ent_ser='0' then
 estado_siguiente<=inicio;
 else
 estado_siguiente<=S_1;
 end if;

 when S_1=>
 if Ent_ser='0' then
 estado_siguiente<=S_10;
 else
 estado_siguiente<=S_1;
 end if;

 when S_10=>
 if Ent_ser='0' then
 estado_siguiente<=inicio;
 else
 estado_siguiente<=S_101;
 end if;

 when S_101=>
```

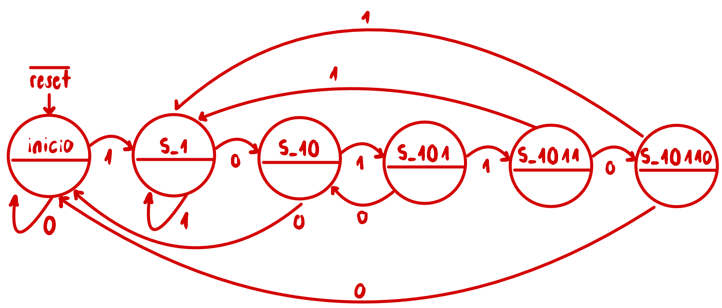
```
if Ent_ser='0' then
estado_siguiente<=S_10;
else
estado_siguiente<=S_1011;
end if;

when S_1011=>
if Ent_ser='0' then
estado_siguiente<=S_10110;
else
estado_siguiente<=S_1;
end if;

when S_10110=>
Det<='1';
if Ent_ser='0' then
estado_siguiente<=inicio;
else
estado_siguiente<=S_1;
end if;

when others=>
estado_siguiente<=inicio;

end case;
end process;
end Behavioral;
```

Diseño 2

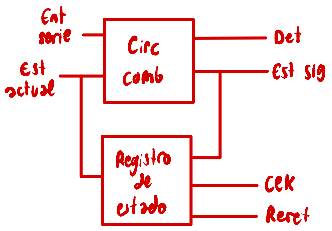
```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
```

Diseño 1

FSM Moore sin solape:

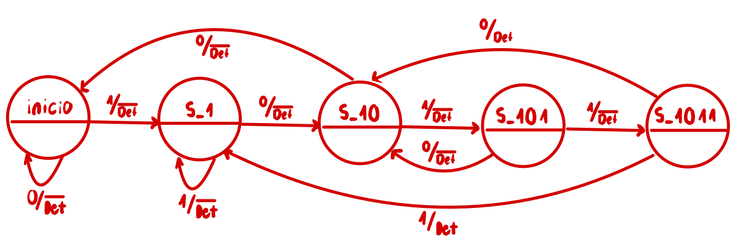


FSM reset asíncrono

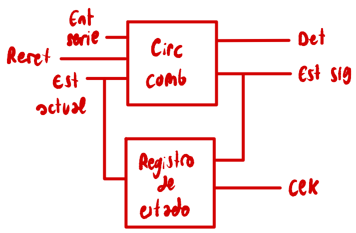


Diseño 2

FSM Mealy con solape:



FSM reset síncrono



```
entity Tema6_problema2 is
   Port ( clk : in  STD_LOGIC;
          reset : in  STD_LOGIC;
          ent_ser : in  STD_LOGIC;
          det : out  STD_LOGIC);
end Tema6_problema2;

architecture Behavioral of Tema6_problema2 is

type estados is (Inicio, S_1, S_10, S_101, S_1011);
signal estado_actual, estado_siguiente: estados;

begin

process (clk)
begin

if rising_edge (clk) then
estado_actual<=estado_siguiente;
end if;
end process;

process (reset,ent_ser,estado_actual)
begin
Det<='0';

if reset='0' then
estado_siguiente<=inicio;

else
```

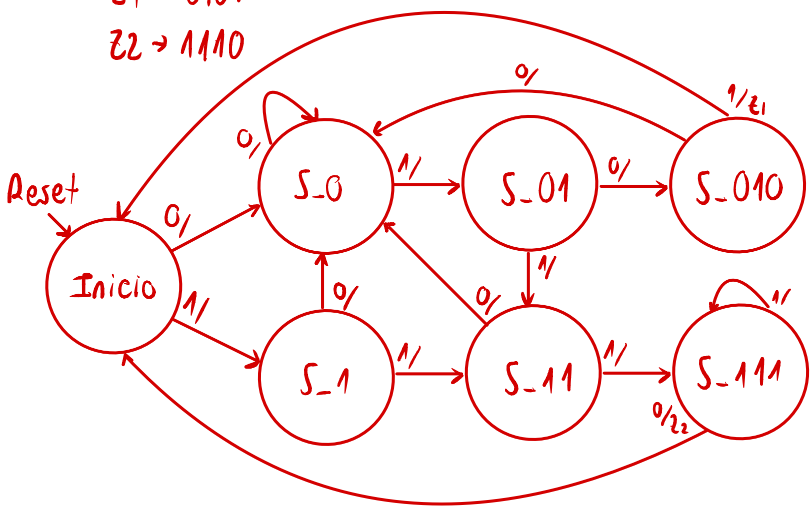
```
case Estado_actual is

when Inicio=>
 if Ent_ser='0' then
 estado_siguiente<=inicio;
 else
 estado_siguiente<=S_1;
 end if;

 when S_1=>
 if Ent_ser='0' then
 estado_siguiente<=S_10;
 else
 estado_siguiente<=S_1;
 end if;

 when S_10=>
 if Ent_ser='0' then
 estado_siguiente<=inicio;
 else
 estado_siguiente<=S_101;
 end if;

 when S_101=>
 if Ent_ser='0' then
 estado_siguiente<=S_10;
 else
 estado_siguiente<=S_1011;
 end if;
```

```
when S_1011=>
if Ent_ser='0' then
estado_siguiente<=S_10;
Det<='1';
else
estado_siguiente<=S_1;
end if;

when others=>
estado_siguiente<=inicio;

end case;

end if;
end process;
end Behavioral;
```

Diseño 3

FSM Mealy
$Z1 \rightarrow 0101$
$Z2 \rightarrow 1110$

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Tema6_problema3 is
   Port ( clk : in  STD_LOGIC;
          reset : in  STD_LOGIC;
          X : in  STD_LOGIC;
          Z1, Z2 : out  STD_LOGIC);
end Tema6_problema3;
```



Reset asíncrono
process (clk, reset)
if reset='1' then nivel alto
if reset='0' then nivel bajo

```
architecture Behavioral of Tema6_problema3 is

type estados is (Inicio, S_0, S_01, S_010, S_1, S_11, S_111);
```

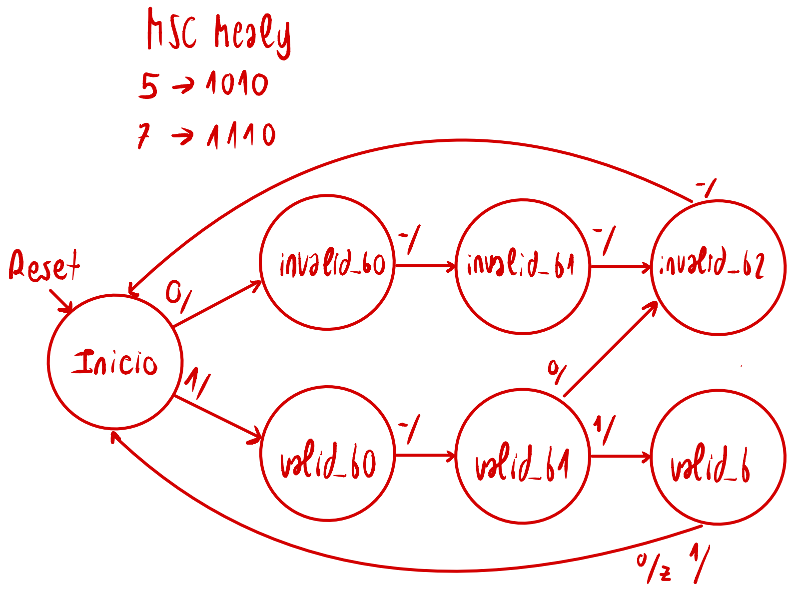
```
signal estado_actual, estado_siguiente: estados;

begin

process (clk,reset)
begin
if reset='0' then
estado_actual<=inicio;
elsif rising_edge (clk) then
estado_actual<=estado_siguiente;
end if;
end process;

process (X,estado_actual)
begin
Z1<='0';
Z2<='0';

case Estado_actual is

when Inicio=>
 if x='0' then
 estado_siguiente<=S_0;
 else
 estado_siguiente<=S_1;
 end if;

 when S_0=>
 if x='0' then
 estado_siguiente<=S_0;
 else
```

```
estado_siguiente<=S_01;
end if;

when S_1=>
if x='0' then
estado_siguiente<=S_0;
else
estado_siguiente<=S_11;
end if;

when S_01=>
if x='0' then
estado_siguiente<=S_010;
else
estado_siguiente<=S_11;
end if;

when S_11=>
if x='0' then
estado_siguiente<=S_0;
else
estado_siguiente<=S_111;
end if;

when S_010=>

if x='0' then
estado_siguiente<=s_0;
else
estado_siguiente<=inicio;
Z1<='1';
```

```
end if;

when S_111=>

if x='0' then
estado_siguiente<=inicio;
Z2<='1';
else
estado_siguiente<=S_111;

end if;

when others=>
estado_siguiente<=inicio;

end case;
end process;
end Behavioral;
```

Diseño 4

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Tema5_problema4 is
   Port ( clk : in  STD_LOGIC;
          reset : in  STD_LOGIC;
          x : in  STD_LOGIC;
          z : out  STD_LOGIC);
end Tema5_problema4;
```




```
architecture Behavioral of Tema5_problema4 is
```

```
type estados is (Inicio, valido_bit0, valido_bit1, valido_bit2, invalido_bit0, invalido_bit1,
invalido_bit2);
signal estado_actual, estado_siguiente: estados;

begin

process (clk)
begin

if rising_edge (clk) then
estado_actual<=estado_siguiente;
end if;
end process;

process (reset,x,estado_actual)
begin
z<='0';

if reset='0' then
estado_siguiente<=inicio;

else

case Estado_actual is

when Inicio=>
 if x='0' then
 estado_siguiente<=invalido_bit0;
 else
 estado_siguiente<=valido_bit0;
 end if;
```

```
when valido_bit0=>
estado_siguiente<=valido_bit1;

when invalido_bit0=>

estado_siguiente<=invalido_bit1;

when valido_bit1=>
if x='0' then
estado_siguiente<=invalido_bit2;
else
estado_siguiente<=Valido_bit2;
end if;

when invalido_bit1=>
estado_siguiente<=invalido_bit2;

when valido_bit2=>
estado_siguiente<=inicio;
if x='0' then
Z<='1';
end if;

when invalido_bit2=>
estado_siguiente<=inicio;

when others=>
estado_siguiente<=inicio;

end case;
```

```
end if;
end process;
end Behavioral;
```

Diseño 6

$2^2 \quad 2^1 \quad 2^0$

$a: \ a_2 \ a_1 \ a_0$

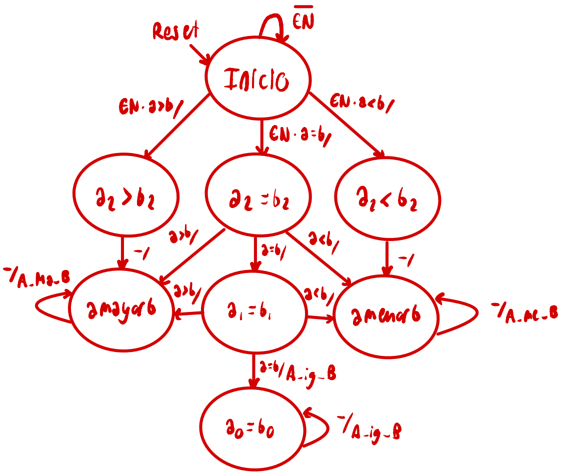
$b: \ b_2 \ b_1 \ b_0$

$a_2 > b_2 \rightarrow a > b$

0 | 1 0

1 | 1 0

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Tema6_problema6 is
   Port ( clk : in  STD_LOGIC;
          reset, a, b, en : in  STD_LOGIC;
          A_ma_B, A_ig_B, A_Me_B : out  STD_LOGIC);
end Tema6_problema6;
```




```
architecture Behavioral of Tema6_problema6 is

type estados is (Inicio, mayor_a2, menor_a2, igual_a2, mayor_a1, menor_a1, igual_a1,
igual_a0);
signal estado_actual, estado_siguiente: estados;

begin

process (clk)
begin

if rising_edge (clk) then
estado_actual<=estado_siguiente;
end if;
end process;

process (EN, a, b,reset,estado_actual)
```

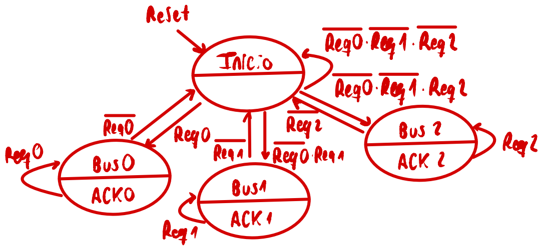
```
begin

a_ma_b<='0';
a_ig_b<='0';
a_me_b<='0';

if reset='1' then

estado_siguiente<=inicio;

else
case Estado_actual is

when Inicio=>
 if EN= '0' then
 estado_siguiente<=inicio;
 else
 if a>b then
 estado_siguiente<=Mayor_a2;
 elsif a=b then
 estado_siguiente<=igual_a2;
 else
 estado_siguiente<=menor_a2;
 end if;
 end if;

 when mayor_a2=>

 estado_siguiente<=mayor_a1;

 when Igual_a2=>
```

```
 if a=b then
  estado_siguiente<=igual_a1;
 elsif a>b then
 estado_siguiente<= mayor_a1;
 elsif a<b then
 estado_siguiente<=menor_a1;
 end if;

 when menor_a2=>
 estado_siguiente<=menor_a1;

 when mayor_a1=>
 estado_siguiente<=mayor_a1;
A_Ma_B<='1';
 when menor_a1=>
 estado_siguiente<=menor_a1;
A_Me_B<='1';
 when igual_a1=>
 if a=b then
 estado_siguiente<=igual_a0;
 elsif a>b then
 estado_siguiente<=mayor_a1;
 A_ma_B<='1';
 elsif a<b then
 estado_siguiente<=menor_a1;
 A_me_B<='1';

  when igual_a0=>
 estado_siguiente<=igual_a0;
A_Ig_B<='1';
end case;
```

```
end if;

end process;
end Behavioral;
```

Diseño 7

FSM árbitro de bus
Moore




```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity Tema6_problema7 is
   Port ( clk : in  STD_LOGIC;
          reset : in  STD_LOGIC;
          req0, req1, req2 : in  STD_LOGIC;
          ack0, ack1, ack2 : out  STD_LOGIC);
end Tema6_problema7;

architecture Behavioral of Tema6_problema7 is

type estados is (Inicio, bus_0, bus_1, bus_2);
signal estado_actual, estado_siguiente: estados;

begin

process (clk, reset)
begin

if reset='0' then
estado_actual<=inicio;
elsif rising_edge (clk) then
estado_actual<=estado_siguiente;
end if;
```

```
end process;

process (req0, req1, req2,estado_actual)
begin

ack0<='0';
ack1<='0';
ack2<='0';

case Estado_actual is

when Inicio=>
 if req0='1' then
 estado_siguiente<=bus_0;
 elsif req1='1' then
 estado_siguiente<=bus_1;
 elsif req2='1' then
 estado_siguiente<=bus_2;
 else
 estado_siguiente<=inicio;
 end if;

 when bus_0=>
 ack0<='1';
 if req0='0'then
 estado_siguiente<=inicio;
 else
 estado_siguiente<=bus_0;
 end if;

 when bus_1=>
```

```
ack1<='1';
if req1='0'then
estado_siguiente<=inicio;
else
estado_siguiente<=bus_1;
end if;

when bus_2=>
ack2<='1';
if req2='0'then
estado_siguiente<=inicio;
else
estado_siguiente<=bus_2;
end if;

end case;

end process;
end Behavioral;
```

<u>Diseño 8 similar a la práctica 4</u>